

Tutorial Questions:

1.
 - a. Explain how the **Active Contour (Snake)** model works in image segmentation.
 - b. Discuss the significance of internal and external forces in the Snake model.
 - c. Write the mathematical formulation of the Snake energy function.

2. A geospatial analysis company is developing a land use classification system to segment satellite images into different regions such as water bodies, forests, and urban areas. They decide to use the Split and Merge technique for segmentation based on pixel intensity variations.
 - a. Explain the working principle of the **Split and Merge technique** in image segmentation.
 - b. What is the condition used for splitting or merging a region? Explain with a suitable example.
 - c. Write the algorithmic steps involved in Split and Merge segmentation.

3. A wildlife conservation project uses drones to monitor animal populations by segmenting different species in aerial images. They employ the **Graph Cuts** method for segmentation.
 - a. Explain the principle of Graph Cuts in image segmentation.
 - b. How does the shortest path algorithm in **Intelligent Scissors** differ from the min-cut/max-flow approach in Graph Cut?
 - c. How does Graph Cut compare with **level set** methods for image segmentation?

4. A robotics team is developing a drone navigation system that uses images from two different positions to estimate 3D structures in the environment using **Structure from Motion (SfM)**.
 - a. Explain what is meant by Structure from Motion (SfM) and Two-Frame SfM.
 - b. Describe how feature points are detected and matched between two frames.
 - c. Explain the process of estimating camera motion and scene structure from the two frames.